SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MANDEM NANDHINI
192472318RC

CO4 - AT2 - Coding Golf Assesment

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QUESTIONS

- **Q1** Bubbled Sorting — 25 marks
- **Q2** Insertion Soring — 25 marks
- **Q3** Quick Sorting — 25 marks
- **Q4** Merger Sort — 25 marks

4/4 answered

Q1 Bubbled Sorting — 25 marks

Write a C code for performing Bubbled Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```c
#include<stdio.h>
int main(){
    int n=5;
    int a[]={8,2,3,5,7};
    int i,j,temp;
    for(i=0;i<n-1;i++){
        for(j=0;j<n-i-1;j++){
            if(a[j]>a[j+1]){
                temp=a[j];
                a[j]=a[j+1];
                a[j+1]=temp;
            }
        }
    }
    for(i=0;i<n;i++){
        printf("%d",a[i]);
    }
    return 0;
}
```

Run | Save | Submit

Input

stdin input

Output

```
23578
```

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4/4 answered

Q2 Insertion Soring — 25 marks

Write a C code for performing Insertion Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```c
#include<stdio.h>
int main(){
    int n=5;
    int a[]={8,2,3,5,7};
    int i,j,key;
    for(i=1;i<n;i++){
        key=a[i];
        j=i-1;
        while(j>0 && a[j]>key){
            a[j+1]=a[j];
            j=j-1;
        }
        a[j+1]=key;
    }
    for(i=0;i<n;i++){
        printf("%d",a[i]);
    }
    return 0;
}
```

Run | Save | Submit

Input

stdin input

Output

```
82357
```

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QUESTIONS

- **Q1** Bubbled Sorting — 25 marks
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- **Q4** Merger Sort — 25 marks

4/4 answered

Q3 Quick Sorting

25 marks

Write a C code for performing Quick Sort - n=7 A[] = {8, 2, 3, 5, 7,1,6}.

Your Code

```c
#include<stdio.h>
void swap(int* a,int* b){
    int t=*a;
    *a=*b;
    *b=t;
}
int partition(int arr[], int low, int high){
    int pivot= arr[high];
    int i =(low-1);
    for(int j=low;j<=high-1;j++){
        if(arr[j]< pivot){
            i++;
            swap(&arr[i],&arr[j]);
            return(i+1);
        }
    }
}
void quickSort(int arr[],int low,int high){
    if(low<high){
        int pi= partition(arr,low,high);
        quickSort(arr,low,pi-1);
        quickSort(arr,pi+1,high);
    }
}
```

Input

stdin input

Output

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QUESTIONS

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- **Q4** Merger Sort — 25 marks

4/4 answered

Q4 Merger Sort

25 marks

Write a C code for performing Merger Sort - n=8 A[] = {8 2 3 5 7 6 4 1}.

Your Code

```c
#include <stdio.h>
void merge(int a[], int l, int m, int r){
    int i=l, j=m+1, k=0, b[8];
    while(i<=m && j<=r)
        b[k++] = (a[i]<a[j]) ? a[i++] : a[j++];
    while(i<=m) b[k++]=a[i++];
    while(j<=r) b[k++]=a[j++];
    for(i=l,k=0;i<=r;i++,k++)
        a[i]=b[k];
}
void sort(int a[], int l, int r){
    if(l<r){
        int m=(l+r)/2;
        sort(a,l,m);
        sort(a,m+1,r);
        merge(a,l,m,r);
    }
}
int main(){
    int a[]={8,2,3,5,7,6,4,1};
    sort(a,0,7);
    for(int i=0;i<8;i++)
        printf("%d ",a[i]);
    return 0;
```

Input

stdin input

Output